TECHNO INDIA UNIVERSITY
WESTBENGAL

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CLASS: BCA 3G

SUBJECT: Fundamentals pf Data Science using R

SUBJECT CODE: TIU-UCA-MJ-L31304

Assignment-1

1. Write an R program to: Read a CSV file into a data frame using read.csv(). Display the first 5 and last 5 rows. Find the mean of a numeric column (e.g., Price) and display summary statistics using summary().

Assignmen-1_Data-Science.R

```r
Name <- c("Rahul", "priyanka", "Aritra", "Akash", "Amit")
Age <- c(20, 21, 19, 22, 20)
Gender <- c("Male", "Female", "Male", "Female", "Male")
Marks <- c(85, 90, 78, 88, 82)

students <- data.frame(Name, Age, Gender, Marks)

write.csv(students, "students_data.csv", row.names = FALSE)

students <- read.csv("students_data.csv")

print("First 5 rows of the data:")
head(students, 5)

print("Last 5 rows of the data:")
tail(students, 5)

mean_marks <- mean(students$Marks)
print(paste("Mean of Marks:", mean_marks))

print("Summary statistics:")
summary(students)
```

21:29 (Top Level)

Console Terminal Background Jobs

R 4.5.3 · ~/

```
> source("~/Assignmen-1_Data-Science.R")
[1] "First 5 rows of the data:"
[1] "Last 5 rows of the data:"
[1] "Mean of Marks: 84.6"
[1] "Summary statistics:"
```

2. Write an R program to create a data frame containing the following information of students: Name, Age, Gender, Marks. Display the complete data frame and the structure of the data frame. Write the data in a CSV file.

Assignmen-1_Data-Science.R

```r
Name <- c("Rahul", "Anita", "Suman", "Priya", "Amit")
Age <- c(20, 21, 19, 22, 20)
Gender <- c("Male", "Female", "Male", "Female", "Male")
Marks <- c(85, 90, 78, 88, 82)

students <- data.frame(Name, Age, Gender, Marks)

print("Students Data Frame:")
print(students)

print("Structure of Data Frame:")
str(students)

write.csv(students, "students_data.csv", row.names = FALSE)
print("Data has been written to students_data.csv")
```

1:1 (Top Level)

Console Terminal Background Jobs

R 4.5.3 · ~/

```
> source("~/Assignmen-1_Data-Science.R")
[1] "Students Data Frame:"
   Name Age Gender Marks
1 Rahul  20   Male    85
2 Anita  21 Female    90
3 Suman  19   Male    78
4 Priya  22 Female    88
5  Amit  20   Male    82
[1] "Structure of Data Frame:"
'data.frame':   5 obs. of  4 variables:
 $ Name  : chr  "Rahul" "Anita" "Suman" "Priya" ...
 $ Age   : num  20 21 19 22 20
 $ Gender: chr  "Male" "Female" "Male" "Female" ...
 $ Marks : num  85 90 78 88 82
[1] "Data has been written to students_data.csv"
```

3. Create a data frame of employees with columns EmpID, Name, Age and Experience. Write an R program to display only those employees who have Experience > 5 years and Age < 40. Write the data in a CSV file.

Assignmen-1_Data-Science.R

Source on Save

```r
EmpID <- c(101, 102, 103, 104, 105)
Name <- c("Rahul", "Anita", "Suman", "Priya", "Amit")
Age <- c(35, 28, 45, 32, 38)
Experience <- c(10, 3, 15, 7, 6)

employees <- data.frame(EmpID, Name, Age, Experience)

print("Employee Data Frame:")
print(employees)

filtered_employees <- employees[employees$Experience > 5 & employees$Age < 40, ]

print("Employees with Experience > 5 years and Age < 40:")
print(filtered_employees)

write.csv(filtered_employees, "filtered_employees.csv", row.names = FALSE)
print("Filtered data written to filtered_employees.csv")
```

12:1 (Top Level)

Console Terminal Background Jobs

R 4.5.3 · ~/

```
> source("~/Assignmen-1_Data-Science.R")
[1] "Employee Data Frame:"
  EmpID  Name Age Experience
1   101 Rahul  35         10
2   102 Anita  28          3
3   103 Suman  45         15
4   104 Priya  32          7
5   105  Amit  38          6
[1] "Employees with Experience > 5 years and Age < 40:"
  EmpID  Name Age Experience
1   101 Rahul  35         10
4   104 Priya  32          7
5   105  Amit  38          6
[1] "Filtered data written to filtered_employees.csv"
```

4. Perform web scraping on a website, extract the data and write the data in a CSV file.

Assignmen-1_Data-Science.R

Source on Save Run

```r
library("rvest")

url = "https://www.amazon.in/s?k=mobile+phone+under+30000&crid=CY3B91R5ATLE&sprefix=mobile+phone+%2Caps%2C446&ref=nb_sb_ss_mvt-tl1-ranker_5_13"

data=read_html(url)

datal=html_nodes(data, '.puis-line-clamp-3-for-col-4-and-8 span , #navbar')
des=html_text(datal)

pricel=html_nodes(data, '.puis-price-instructions-style .a-price-whole , #navbar')
price=html_text(pricel)

df=data.frame(DESCRIPTION=des, PRICE=price)

print(df)

write.csv(df, "Phones.csv")
```

2	Samsung Galaxy M56 5G Mobile (Light Green,8GB RAM,256GB Storage)| Segment's Slimmest|Gorilla Glass Victus+| 10 Bit HDR Video| 50MP Camera| AI | Vapour Cooling Chamber| Lag-free Gaming| Without Charger	24,999
3	Samsung Galaxy S26 5G (Black, 12GB RAM, 256GB Storage), AI Phone, Photo Assist, Creative Studio, 50MP Camera with ProVisual Engine, Powerful Customized Processor and 4300mAh Battery	87,999
4	vivo T4 5G (Emerald Blaze, 12GB RAM+256GB Storage)| Snapdragon 7s Gen 3 | 120Hz Quad-Curved AMOLED | 7300mAh+90W Fast Charge | 50MP Sony OIS Camera | Military Grade Build | AI Photo Enhance | AI Erase	28,314
5	iQOO Z10R 5G (Aquamarine, 12GB RAM, 256GB Storage) | 32MP 4K Selfie Camera | Quad-Curved AMOLED Display | Dimensity 7400 Processor with 750K+ AnTuTu Score	26,998
6	iQOO Neo 10R 5G (Raging Blue, 8GB RAM, 256GB Storage) | Snapdragon 8s Gen 3 Processor | India's Slimmest 6400mAh Battery Smartphone | Segment's Most Stable 90FPS for 5 Hours	28,998
7	iQOO Z11x 5G (Titan Black, 8GB RAM, 256 GB Storage) | Dimensity 7400-Turbo Processor | 7200 mAh Battery Smartphone | Powered by OriginOS 6	22,998
8	OnePlus Nord CE5 | MediaTek Dimensity | Massive 7100mAh Battery | Powered by OnePlus AI | 128GB 8GB | Marble Mist	24,996
9	vivo Y400 Pro 5G (Nebula Purple, 8GB RAM, 256GB Storage) with No Cost EMI/Additional Exchange Offers	27,999
10	iQOO Z11x 5G (Prismatic Green, 8GB RAM, 256 GB Storage) | Dimensity 7400-Turbo Processor | 7200 mAh Battery Smartphone | Powered by OriginOS 6	22,998
11	vivo T4R 5G (Twilight Blue, 256 GB) (12 GB RAM)	26,074.00
12	realme 15 Pro 5G Smartphone 8+256GB,4K 60FPS,144Hz HyperGlow 4D Curve Display,7000mAh Battery,80W SuperVOOC, Camera Triple 50MP, Snapdragon 7 Gen 4, IP69, Purple	37,999
13	REDMI Note 15 Pro+ 5G (Carbon Black, 8GB + 256GB) | 200MasterPixel OIS Camera | Snapdragon 7s Gen 4 | CrystalRes AMOLED | IP69/69K | 6500mAh Si/C Battery | 100W HyperCharge	37,999
14	Samsung Galaxy M56 5G Mobile (Black, 8GB RAM, 256GB Storage)| Segment's Slimmest | Gorilla Glass Victus+| 10 Bit HDR Video| 50MP Camera| AI | Vapour Cooling Chamber| Lag-Free Gaming| Without Charger	24,999
15	Motorola Edge 60 Fusion 5G (Pantone Mykonos Blue, 12GB RAM, 256GB Storage)	24,499
16	Google Pixel 7 5G (Obsidian, 8GB RAM, 128GB Storage)	29,999
17	realme NARZO 80 Pro 5G (Speed Silver,12GB+256GB) | Segment's 1st MediaTek Dimensity 7400 Chipset | 6000mAh Titan Battery + 80W Ultra Charge | 4500nits HyperGlow Esports Display | IP69 Waterproof	21,999
18	Motorola Edge 70 (Grey, 8GB RAM, 256GB Storage)	29,831
19	realme NARZO Power 5G (Titan Silver,8GB+256GB) 10001mAh Power. Stay Unstoppable. | 144Hz Curved+ AMOLED Display | Revolutionary Dual-Chip System | 50MP Sony OIS Camera	29,999
20	Motorola Edge 60 Fusion 5G (Pantone Slipstream, 12GB RAM, 256GB Storage) | Dimensity 7400 | 5500mAh Battery | 68W Fast Charge | 6.67寰ｸ pOLED | 50MP+13MP | 32MP Front | Gorilla Glass 7i	25,999
21	REDMI Note 15 5G (Black, 8GB RAM 256GB Storage) | 108MP OIS Camera | Snapdragon 6 Gen 3 | 17.2cm Tough Curved AMOLED Screen | Segment寰冱 Slimmest Phone | 5520mAh Si/C Battery | 45W Fast Charging	24,998

5.Write a program to collect mobile description and price under 30000 from Flipkart website.

```r
library("rvest")

url = "https://www.amazon.in/s?k=mobile+phone+under+30000&crid=CY3B91R5ATLE&sprefix=mobile+phone+%2Caps%2C446&ref=nb_sb_ss_mvt-tl1-ranker_5_13"

data=read_html(url)

datal=html_nodes(data, '.puis-line-clamp-3-for-col-4-and-8 span , #navbar')
des=html_text(datal)

pricel=html_nodes(data, '.puis-price-instructions-style .a-price-whole , #navbar')
price=html_text(pricel)

df=data.frame(DESCRIPTION=des, PRICE=price)

write.csv(df, "Phones.csv")
```

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3	Samsung Galaxy S26 5G (Black, 12GB RAM, 256GB Storage), AI Phone, Photo Assist, Creative Studio, 50MP Camera with ProVisual Engine, Powerful Customized Processor and 4300mAh Battery	87,999
4	vivo T4 5G (Emerald Blaze, 12GB RAM+256GB Storage)| Snapdragon 7s Gen 3 | 120Hz Quad-Curved AMOLED | 7300mAh+90W Fast Charge | 50MP Sony OIS Camera | Military Grade Build | AI Photo Enhance | AI Erase	28,314
5	iQOO Z11x 5G (Titan Black, 8GB RAM, 256 GB Storage) | Dimensity 7400-Turbo Processor | 7200 mAh Battery Smartphone | Powered by OriginOS 6	22,998
6	iQOO Neo 10R 5G (Raging Blue, 8GB RAM, 256GB Storage) | Snapdragon 8s Gen 3 Processor | India's Slimmest 6400mAh Battery Smartphone | Segment's Most Stable 90FPS for 5 Hours	28,998
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10	OnePlus Nord CE5 | MediaTek Dimensity | Massive 7100mAh Battery | Powered by OnePlus AI | 128GB 8GB | Marble Mist	24,996
11	vivo Y400 Pro 5G (Nebula Purple, 8GB RAM, 256GB Storage) with No Cost EMI/Additional Exchange Offers	27,999
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15	Samsung Galaxy M56 5G Mobile (Black, 8GB RAM, 256GB Storage)| Segment's Slimmest | Gorilla Glass Victus+| 10 Bit HDR Video| 50MP Camera| AI | Vapour Cooling Chamber| Lag-Free Gaming| Without Charger	24,999
16	REDMI Note 15 5G (Black, 8GB RAM 256GB Storage) | 108MP OIS Camera | Snapdragon 6 Gen 3 | 17.2cm Tough Curved AMOLED Screen | Segment宴沍 Slimmest Phone | 5520mAh Si/C Battery | 45W Fast Charging	24,998
17	Motorola Edge 70 (Grey, 8GB RAM, 256GB Storage)	29,830
18	realme NARZO 80 Pro 5G (Speed Silver,12GB+256GB) | Segment's 1st MediaTek Dimensity 7400 Chipset | 6000mAh Titan Battery + 80W Ultra Charge | 4500nits HyperGlow Esports Display | IP69 Waterproof	21,999
19	Samsung Galaxy A55 5G (Awesome Navy, 12GB RAM, 256GB Storage) | AI Mobile | 50MP Main Camera (OIS) | Super HDR Video| Nightography | IP67 | Gorilla Glass Victus+ | sAMOLED | Gaming Smartphone	29,999
20	Motorola Edge 60 Fusion 5G (Pantone Slipstream, 12GB RAM, 256GB Storage) | Dimensity 7400 | 5500mAh Battery | 68W Fast Charge | 6.67宴ｳ pOLED | 50MP+13MP | 32MP Front | Gorilla Glass 7i	25,999
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